

Practice Problems Set 1: Complex Numbers

Problem 1

- If $z = 5$, plot z and jz on an Argand diagram. Deduce that multiplying z by j causes an anticlockwise rotation through 90° about the origin.
- If $z = 2 + 3j$, plot z and jz . Show that the same behaviour as in part (a) occurs. What can you deduce about multiplying a complex number by j , in general?

Problem 2: Express $\frac{1-j}{1+j}$ in the form $a + bj$ via two methods: one using the Cartesian form throughout, and one changing numerator and denominator to exponential form. Show that the two answers agree.

Problem 3: Calculate each of the following two ways: first by using the binomial theorem and second by changing to exponential form and using DeMoivre's formula.

- $(1-j)^4$
- $(1+j\sqrt{3})^3$

Problem 4: Make a table with four columns. Each row will contain four representations of a complex number z : the Cartesian expression $z = a + jb$; the polar expression, $z = r(\cos \theta + j \sin \theta)$; and the exponential representation $z = re^{j\theta}$; and a little picture of the complex plane with the complex number marked on it. There are five rows, containing, in one column or another, the following complex numbers:

- $1-j$
- z such that $|z| = 2$ and $\text{Arg}(z) = \frac{\pi}{6}$
- A square root of j with negative real and imaginary parts
- A sixth root of 1 with argument θ such that $0 < \theta < \frac{\pi}{2}$
- $\left[(1+j)/\sqrt{2} \right]^{-13}$

Problem 5: Find the complex roots of the equation: $x^4 + 16 = 0$

Problem 6: Show that

$$\text{a) } \cos \theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta}) \quad \text{b) } \sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$$

Problem 7:

- Find non-negative real numbers A , ω , and ϕ such that $\text{Re}\left(\frac{je^{2jt}}{1+j}\right) = A\cos(\omega t + \phi)$
- Express the cube roots of $8j$ in the form $a + bj$